SOC HOST TRIAGE REPORT

Chapter 8 Mini Lab — Linux Security Investigation (Kali Linux)

AUTHOR	LAB DATE	TARGET OS	SCOPE
[Your Name]	22 October 2026	Kali Linux amd64	Host Baseline & Triage

1. EXECUTIVE SUMMARY & LAB OVERVIEW

This report documents a beginner-level SOC host investigation executed on a Kali Linux virtual machine. Acting as a junior SOC analyst, initial host triage was conducted to assess system identity, running software, memory usage, network interface exposure, system logs, and firewall posture. All 18 terminal executions performed were strictly read-only; no system configurations, services, or firewall rules were modified.

2. TARGET ENVIRONMENT SPECIFICATIONS

Parameter	Value	Source / Verification
Operating System	Kali Linux (Rolling)	Prompt format `kali@kali`, kernel string
Kernel Version	6.19.14+kali-amd64	`uname -r` output
Host & User Identity	kali / kali	`hostname` and `whoami`
Desktop / Shell	Xfce4 / zsh / QTerminal	Process table inspection (`ps aux`)
Virtualization Platform	VirtualBox Guest	Guest services (`VBoxClient`, `VBoxDRMClient`)
Privileges	Standard user with sudo rights	Sudo password prompts during execution

3. DETAILED COMMAND LOGS & ANALYSIS

Task 1 — System Identification

Host identity verification establishes the target baseline for incident attribution and inventory correlation.

```
$ whoami    READ-ONLY
```

Output: `kali` — Confirms effective user context. Starts investigation without root privileges.

```
$ hostname    READ-ONLY
```

Output: `kali` — Ties system log markers (e.g. journal entries) to this target host.

```
$ uname -r    READ-ONLY
```

Output: `6.19.14+kali-amd64` — Kernel architecture and patch level required for vulnerability checks.

Task 2 — Process Baseline & Analysis (CPU)

```
$ ps aux --sort=-%cpu | head -n 10    READ-ONLY
```

Analysis: Highest CPU consumption belonged to `Xorg` (0.8% CPU, PID 741) and VirtualBox guest helpers (`VBoxClient`). No anomalous background CPU activity detected. Command-line parameters for Xorg show `-nolisten tcp`, indicating X11 is isolated locally.

EVIDENCE CAPTURE 1 — IDENTIFICATION & PROCESS SNAPSHOTS

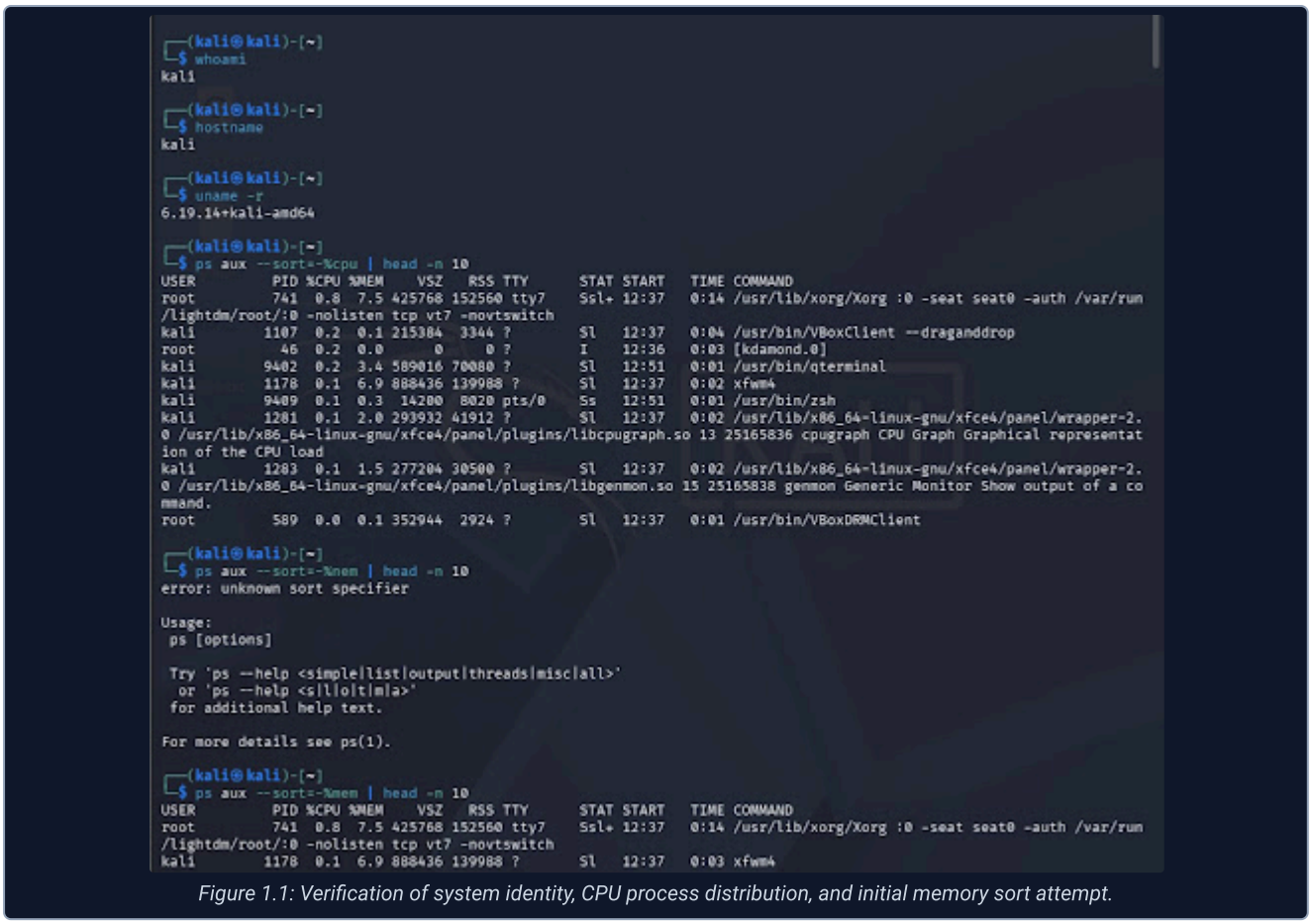


Figure 1.1: Verification of system identity, CPU process distribution, and initial memory sort attempt.

Task 2 — Process Analysis (Memory) & Sockets

```
$ ps aux --sort=-%mem | head -n 10    READ-ONLY
```

Analysis: Top RAM usage is dominated by the Xorg display server (7.5% RSS) and Xfce desktop environment components (`xfwm4` at 6.9%, `blueman-applet` at 3.1%, `xfdesktop` at 2.9%). Total system footprint is light and representative of a default VM environment with no memory-leak or mining behavior.

Task 3 — Listening Network Sockets

```
$ sudo ss -tulpn    READ-ONLY
```

Analysis: The execution yielded **zero active listening TCP or UDP sockets**. The network attack surface is completely closed at the host layer in this state.

Socket Question	Finding	SOC Implication
Active Listeners?	None	No unauthenticated inbound exposure
Bound Ports?	N/A	No web, SSH, or RPC services open
Exposure Rule	Does bound port = Internet access?	**No.** Port visibility relies on bound interface (127.0.0.1 vs 0.0.0.0), network NAT, and local/gateway firewalls.

EVIDENCE CAPTURE 2 — MEMORY PROFILE, SOCKETS & RUNNING SERVICES

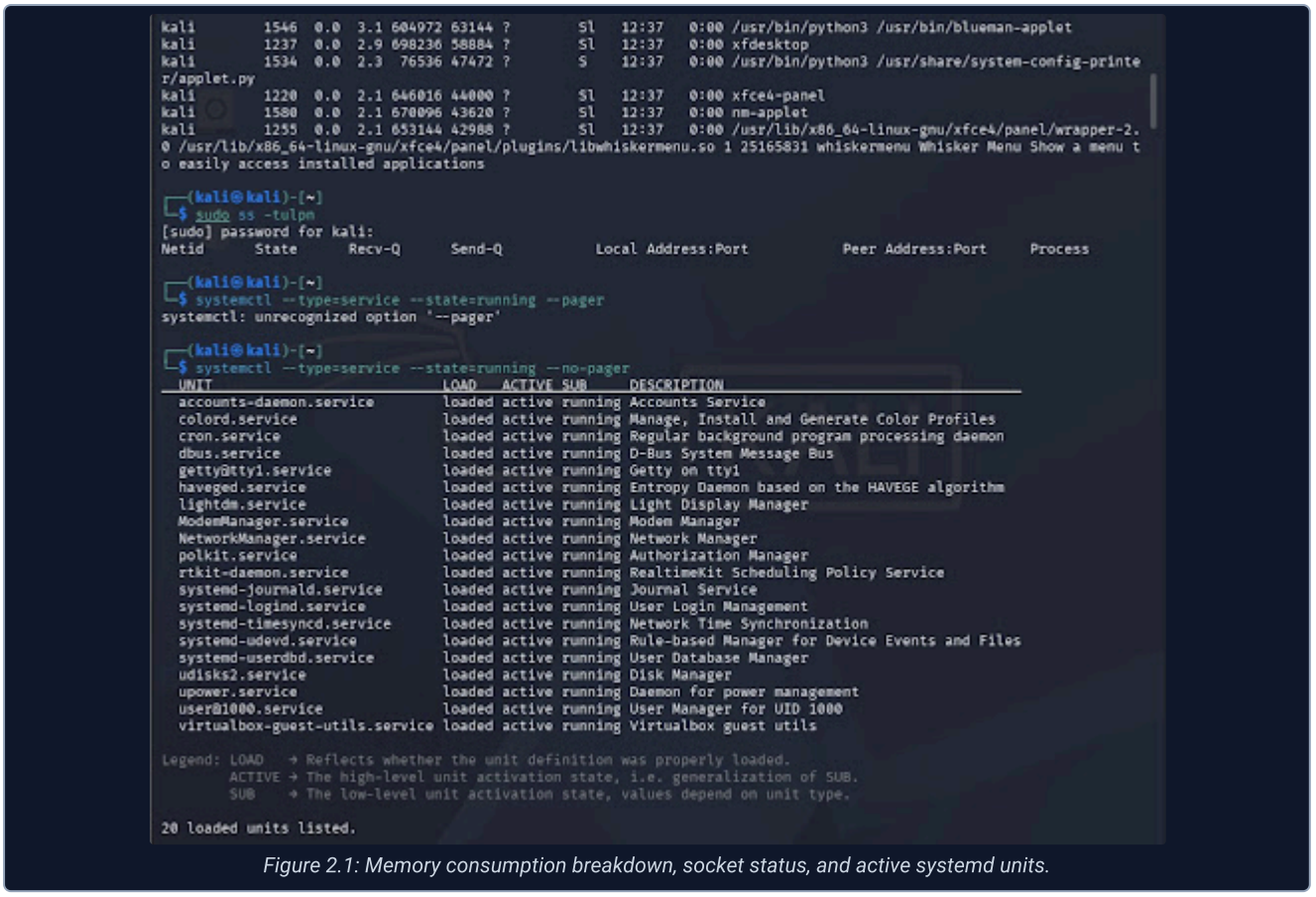


Figure 2.1: Memory consumption breakdown, socket status, and active systemd units.

Task 4 — System Services Inspection

```
$ systemctl --type=service --state=running --no-pager    READ-ONLY
```

Analysis: 20 service units were reported active. Core services include `NetworkManager`, `systemd-journald`, `lightdm`, and `virtualbox-guest-utils`. No server daemons (e.g., `apache2`, `nginx`, `smbd`) were running.

```
$ systemctl status ssh --no-pager    READ-ONLY
```

Output: `Loaded: loaded (... disabled)|Active: inactive (dead)`.
SOC Takeaway: OpenSSH binary is installed on disk but service is disabled at boot and inactive in memory.

Task 5 — Journal Error Analysis

```
$ journalctl -b -p err -n 20 --no-pager    READ-ONLY
```

Logged Errors: 7 error-level messages recorded. Analysis shows DRM graphics driver incompatibility (`vmwgfx` on VirtualBox hypervisor), missing `gkr-pam` control file, and missing Evolution data server unit for Bluetooth `obexd`. No authentication anomalies or exploit indicators observed.

EVIDENCE CAPTURE 3 — SSH STATUS, JOURNAL ERRORS & NFTABLES INSPECTION

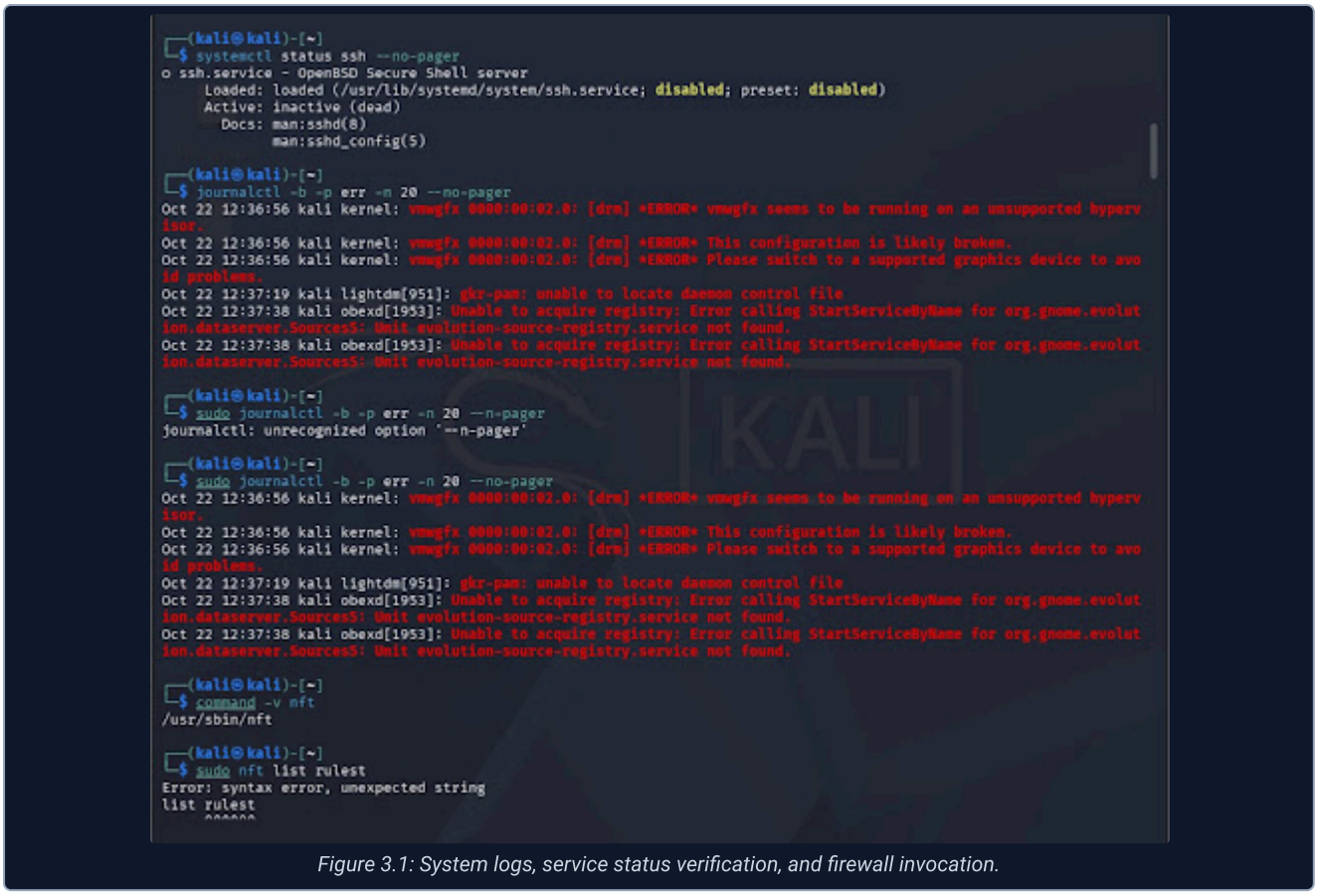


Figure 3.1: System logs, service status verification, and firewall invocation.

Task 6 — Firewall Ruleset Inspection (nftables View)

```
$ sudo nft list ruleset    READ-ONLY
```

Rule Engine Analysis: Output reveals rules managed via the `iptables-nft` translation bridge and structured around UFW (Uncomplicated Firewall) chains:

- **INPUT Policy:** Set to DROP. Inbound connections default-deny.
- **OUTPUT Policy:** Set to ACCEPT. Outbound connections allowed.
- **Allowed Inbound:** Loopback traffic (`lo`) and established/related state connections.
- **User Rules:** Specific entry accepting TCP port 445 from subnet `10.0.2.0/24`.
- **Logging & Counters:** Logging chain `[UFW BLOCK]` registered 1 packet (635 bytes) dropped under multicast evaluation (`ufw-not-local`).

EVIDENCE CAPTURE 4 — IPTABLES VIEW & FIREWALL COUNTERS

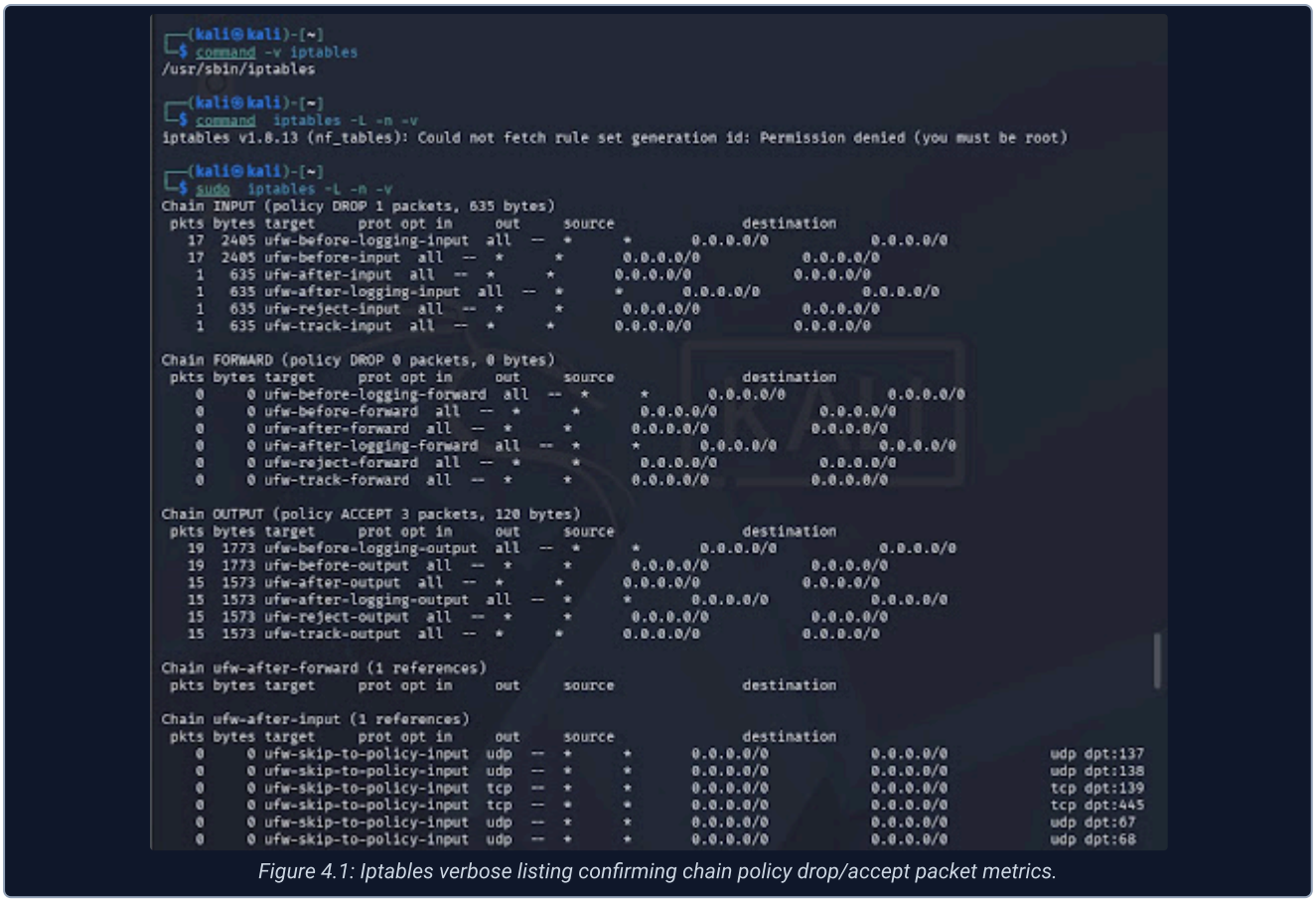


Figure 4.1: Iptables verbose listing confirming chain policy drop/accept packet metrics.

```
$ sudo iptables -L -n -v    READ-ONLY
```

Cross-verification via iptables interface corroborates nftables data: `Chain INPUT (policy DROP 1 packets, 635 bytes)`. Chain `ufw-user-input` verifies the single explicit SMB rule (TCP 445) targeting `10.0.2.0/24` with 0 matching hits.

4. FINAL CHALLENGE ASSESSMENT

#	Investigation Core Question	Analyst Conclusion
1	What does `ss -tulpn` investigate?	Identifies listening TCP/UDP sockets, associated ports, and process IDs.
2	Does a listening port prove external exposure?	No. Interface binding, local firewall rules, and NAT gateways restrict exposure.
3	What primary role does `journalctl` fill?	Querying structured system logs maintained by the systemd-journald daemon.
4	First triage step for an unfamiliar process?	Examine process executable path, owner UID, parent process PID, and command arguments.

5. TROUBLESHOOTING & SYNTAX LOG

Target Command	Error Encountered	Root Cause	Corrective Action	Outcome
`ps aux --sort=-%mem`	Unknown sort specifier	Character syntax typo	Re-typed exact flag	**Resolved**
`systemctl ... --pager`	Unrecognized option	Invalid flag name	Switched to `--no-pager`	**Resolved**
`journalctl ... --n-`	Unrecognized option	Typo in flag double-dash	Corrected to `--no-`	**Resolved**